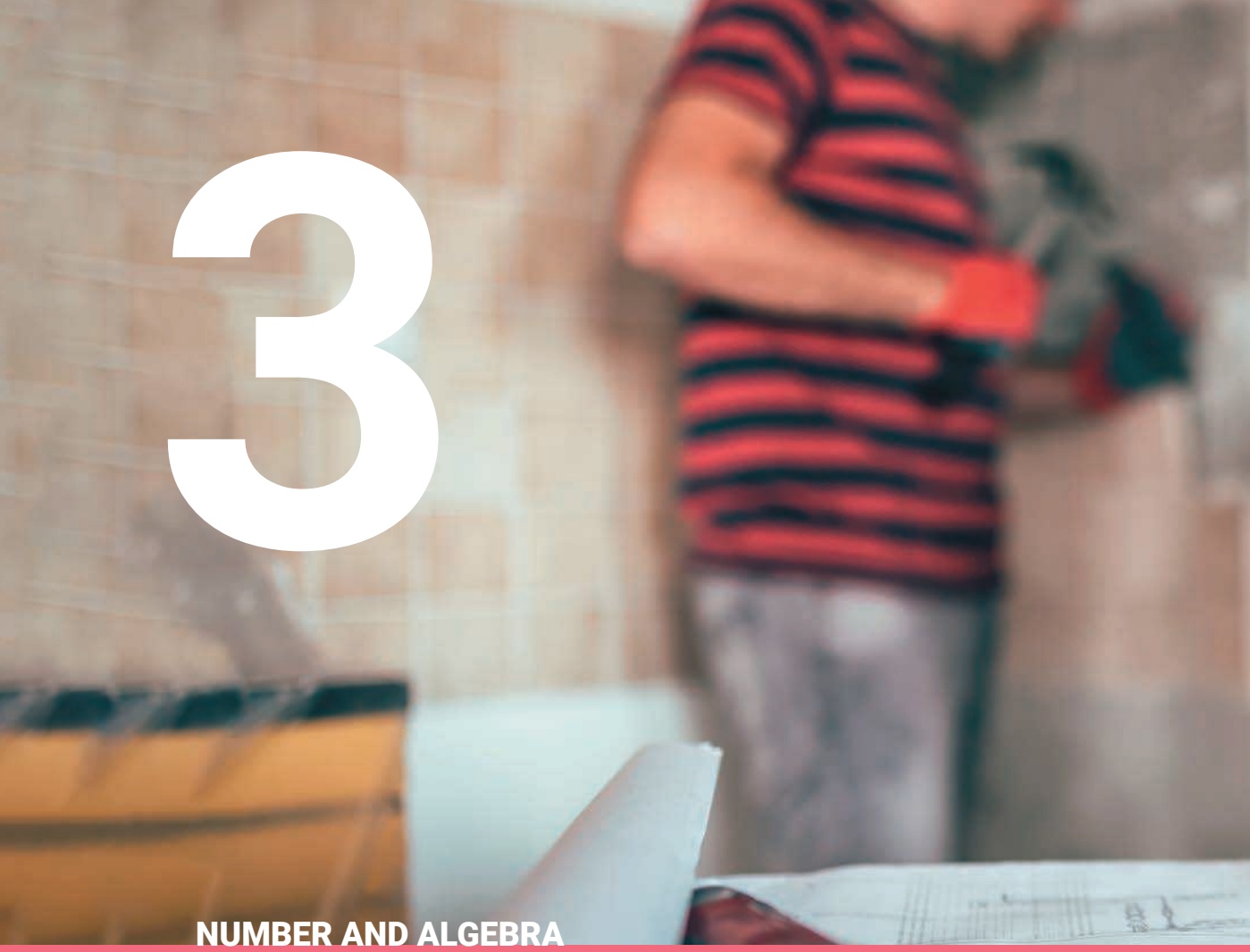


3

NUMBER AND ALGEBRA

NUMERACY AND CALCULATION

Numbers are an important part of our lives. We work with numbers when we shop for the best deal and when we borrow or invest money. Calculations are made when using recipes to cook and in the design and construction of houses. Whenever numbers are used to help us solve problems in our lives, calculations will need to be made.



Chapter outline

	Working mathematically				
3.01 Recurring decimals	U	F	PS	R	C
3.02 Applying percentages	U	F	PS	R	C
3.03 Profit, discounts and GST	U	F	PS	R	
3.04 Simple interest	U	F	PS	R	
3.05 Ratios and rates	U	F	PS	R	
3.06 Converting rates	U	F	PS	R	

Wordbank

cost price The price an item costs the retailer

GST Goods and services tax, a 10% tax added to the original price of an item or service

loss The amount lost when selling an item at a lower price

per annum (p.a.) Per year

principal An amount of money invested or borrowed, on which interest is given or charged

profit The amount made when selling an item at a higher price

recurring decimal A decimal with one or more digits that repeat endlessly, for example, $0.16 = 0.1666\ldots$

unitary method A method of finding a quantity by finding one part or 1% first

Quiz
Wordbank 3

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (10):

3.02 Percentage increase and decrease • The unitary method

3.03 Profit and loss • Discounts and GST • GST

3.04 Simple interest

3.05 Ratio problems • Dividing a quantity in a given ratio • Rate problems

3.06 Converting rates 2

Twig video (1):

SkillCheck Chinese development of maths

PhET interactives (3):

3.05 Ratio and proportion • Proportion playground • Unit rates

Quizzes (6):

- Wordbank 3
- SkillCheck 3
- Mental skills 3A
- Mental skills 3B
- Language of maths 3
- Test yourself 3

Skillsheets (6):

SkillCheck Integers • Integers using diagrams • Decimals

3.02 Fractions, decimals and percentages • Mental percentages

3.05 Ratios

Worksheets (8):

SkillCheck Where's the point?

3.01 Fraction families

3.02 Working with percentages

3.03 Profit and loss • Discounts and special offers

3.04 Simple interest

3.05 Rates exercise

Mind map Numeracy and calculation (Advanced)

Puzzles (5):

3.02 Percentages to fractions • Fractions to percentages • Percentages of an amount

3.03 Percentage profit and loss

Language of maths Numbers crossword

Technology (3):

3.01 Terminating and recurring decimals

3.02 Percentages, fractions and decimals • Converting marks to percentages

Spreadsheet (1):

3.02 Percentages, fractions and decimals

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- convert between fractions and decimals
- (PATH) convert recurring decimals to fractions
- solve problems involving percentages, ratios and rates
- solve problems involving profit, loss, discounts, GST and simple interest
- convert between units for rates, for example, kilometres per hour to metres per second

SkillCheck ANSWERS ON P. 615

1 Evaluate each expression.

a	$7 \times (-8)$	**b**	$-120 \div (-6)$	**c**	$7 - (-15)$
d	$-10 \times 5 \times (-2)$	**e**	$\frac{27}{-3}$	**f**	$20 - 5 \times 8$
g	$(-3)^2 + (-2)^2$	**h**	$(-10 + 7) \times 4$	**i**	$-24 \div 6 \times (-5)$
j	$\frac{-100}{-5} \times 4$	**k**	$-8 - (10 - 15)$	**l**	$(-4)^3$
m	$6 - (-30) \div 6$	**n**	$40 + 8 \times (-9)$	**o**	$(-5)^2 + 4 \times (-7)$
p	$24 \div (-6) + (-5) \times 3$	**q**	$12 - (4 \times 8)$	**r**	$64 \div (-4)^2$

2 Robyn had to calculate $36.024 \div 0.08$ without using a calculator. Which of these calculations will give her the same answer? Select **A**, **B**, **C** or **D**.

A $36\,024 \div 8$ **B** $36.024 \div 8$ **C** $3602.4 \div 8$ **D** $0.360\,24 \div 8$

3 Evaluate each expression.

a	$\frac{2}{3} + \frac{4}{7}$	**b**	$\frac{3}{5} + \frac{1}{2}$	**c**	$\frac{5}{9} - \frac{1}{4}$	**d**	$\frac{9}{10} - \frac{2}{3}$
e	$\frac{5}{7} \times \frac{3}{4}$	**f**	$\frac{1}{2} \times \frac{3}{5}$	**g**	$6 \times \frac{3}{4}$	**h**	$\frac{2}{3} \times 7$
i	$\frac{3}{4} \div \frac{1}{2}$	**j**	$\frac{2}{3} \div \frac{4}{5}$	**k**	$\frac{3}{7} \div 5$	**l**	$\frac{6}{7} \div \frac{2}{3}$

4 Express each number as a percentage.

a	$\frac{1}{4}$	**b**	0.3	**c**	0.07	**d**	$\frac{3}{5}$

Video Chinese development of maths

Quiz SkillCheck 3

Skillsheets Integers
Integers using diagrams
Decimals

Worksheet Where's the point?

Recurring decimals 3.01

Recurring decimals have a pattern of numbers that repeat or recur. We use dots to indicate the digits that repeat. Here are some examples:

$0.4444... = 0.\dot{4}$

$0.588\,88... = 0.5\dot{8}$

$0.636\,363... = 0.\dot{6}\dot{3}$

$0.415\,415\,415... = 0.\dot{4}1\dot{5}$

Terminating decimals do not repeat (terminating means 'stopping'). Here are some examples:

0.6 0.875 0.057

Worksheet Fraction families

Technology Terminating and recurring decimals

To convert a fraction to a decimal, divide its **numerator** by its **denominator**.

Example 1

Convert each fraction to a decimal and state whether it is recurring or terminating.

a $\frac{5}{6}$ **b** $\frac{11}{40}$

Solution

a $\frac{5}{6} = 5 \div 6$
$= 0.8333...$
$= 0.8\dot{3}$ (a recurring decimal)

b $\frac{11}{40} = 11 \div 40$
$= 0.275$ (a terminating decimal)

Because both terminating and recurring decimals can be written as fractions, they are both examples of **rational numbers**.

Some decimals are neither recurring nor terminating. They have an infinite number of digits but there is no pattern. These are **irrational numbers**. Here are some examples:

$\pi = 3.141\ 592\ 653...$ $\sqrt{3} = 1.732\ 050\ 808...$

The rational and irrational numbers together make up the **real numbers**.

PATH

Converting recurring decimals to fractions

A recurring decimal can be converted to a fraction using an algebraic method.

Example 2

Convert each recurring decimal to a simple fraction.

a $0.1\dot{7}$ **b** $0.\dot{5}\dot{4}$

Solution

a Let $x = 0.1\dot{7}$
$= 0.177777...$
$\therefore 10x = 1.77777...$ One digit is recurring, so multiply by 10. This deletes the recurring digits.
$\therefore 10x - x = 1.77777... - 0.177777...$
$9x = 1.6$
$x = \frac{1.6}{9}$
$= \frac{16}{90}$ Multiplying numerator and denominator by 10.
$= \frac{8}{45}$

Since $x = 0.1\dot{7}$, $0.1\dot{7} = \frac{8}{45}$

This answer can be checked by converting $\frac{8}{45}$ to a decimal on a calculator:

$\frac{8}{45} = 0.177777...$

On some calculators, you can enter many digits of a recurring decimal and it will convert it into a fraction. For example:

0.17777777777777777777777777 = S⇔D.

The S⇔D key converts between standard fractions and decimals.

b Let $x = 0.\dot{5}\dot{4}$

$= 0.545454...$

$\therefore 100x = 54.545454...$ 2 digits are recurring, so multiply by 100.

$\therefore 100x - x = 54.545454 .... - 0.545454...$ This deletes the recurring digits.

$99x = 54$

$x = \frac{54}{99}$

$= \frac{6}{11}$

Since $x = 0.\dot{5}\dot{4}$, $0.\dot{5}\dot{4} = \frac{6}{11}$

Checking:

$\frac{6}{11} = 0.545454 ....$

Or converting by calculator:

0.545454545454545454 = S⇔D.

PATH

3.01

EXERCISE 3.01 ANSWERS ON P. 615

Recurring decimals U F PS R C

1 Write each recurring decimal using dot notation. C

a 0.166 666... **b** 0.273 273 27... **c** 0.888 88... **d** 0.209 99...

2 Show the repeating pattern for each recurring decimal. C

a $0.\dot{1}09\dot{8}$ **b** $0.1\dot{7}$ **c** $0.2\dot{5}$ **d** $8.3\dot{1}\dot{8}$

3 Convert each fraction to a decimal and state whether it is recurring or terminating. C

EXAMPLE 1

a $\frac{7}{8}$ **b** $\frac{3}{11}$ **c** $4\frac{8}{9}$ **d** $\frac{13}{5}$ **e** $\frac{19}{25}$

f $\frac{2}{3}$ **g** $\frac{13}{20}$ **h** $1\frac{7}{12}$ **i** $8\frac{13}{40}$ **j** $\frac{17}{24}$

4 **a** Convert $\frac{1}{3}$ to a recurring decimal. C

b Use your answer to part **a** to determine the value of: R

i $0.\dot{6}$ **ii** $0.\dot{9}$

5 Convert each recurring decimal to a simple fraction. C

PATH

EXAMPLE 2

a $0.\dot{2}$ **b** $0.\dot{8}$ **c** $0.1\dot{5}$ **d** $0.3\dot{8}$

e $0.4\dot{6}$ **f** $0.6\dot{5}$ **g** $0.3\dot{1}$ **h** $0.62\dot{4}$

i $0.58\dot{3}$ **j** $0.8\dot{3}2\dot{2}$ **k** $0.0\dot{3}$ **l** $0.7\dot{5}$

6 **a** Convert $0.4\dot{5}$ to a simple fraction.

b Use your answer from part **a** to write $0.\dot{9}\dot{0}$ as a simple fraction. R C

7 **a** Convert $0.6\dot{3}\dot{4}$ to a simple fraction.

b Use your answer from part **a** to write $3.6\dot{1}\dot{5}$ as a simple fraction. PS R C

8 **a** Convert $2.04\dot{5}$ to a simple fraction.

b Use your answer from part **a** to write $7.08\dot{5}$ as a simple fraction. PS R C

Foundation Standard Complex

3.02 Applying percentages

Skillsheets
Fractions, decimals and percentages

Mental percentages

Worksheet
Working with percentages

Puzzles
Percentages to fractions

Percentages of an amount

Fractions to percentages

Video
Percentage increase and decrease

Example 3

Find:

a 12% of \$90 **b** $5\frac{1}{2}$% of \$540

Solution

a $12\% \times \$90 = \frac{12}{100} \times \$90$ or $0.12 \times \$90$

$= \$10.80$

Entering the decimal 0.12 for 12% on a calculator is faster

b $5\frac{1}{2}\%$ of $\$540 = 5\frac{1}{2} \div 100 \times \$540$ or $0.055 \times \$540 = \$29.70$

Example 4

a Increase \$200 by 7%. **b** Decrease \$150 by 12%.

Solution

a Increase $= 7\% \times \$200$

$= \$14$

$\therefore$ New amount $= \$200 + \$14$

$= \$214$

$100\% + 7\% = 107\%$

or New amount $= 107\%$ of \$200

$= 1.07 \times \$200$

$= \$214$

b Decrease $= 12\% \times \$150$

$= 18$

$\therefore$ New amount $= \$150 - \$18$

$= \$132$

$100\% - 12\% = 88\%$

or New amount $= 88\%$ of \$150

$= 0.88 \times \$150$

$= \$132$

Technology
Percentages, fractions and decimals

Converting marks to percentages

Spreadsheet
Percentages, fractions and decimals

Expressing quantities as percentages

To **express one quantity as a percentage of another**, write the appropriate fraction and then change it to a percentage:

$$\text{Percentage} = \frac{\text{amount}}{\text{whole amount}} \times 100\%$$

or Percentage = amount ÷ whole amount × 100%

Foundation Standard Complex

9780170479585

Example 5

a Jamila scores 43 out of 60 in a History test. What is her percentage mark to the nearest whole number?

b What percentage (correct to one decimal place) is 35 s of 5 min?

Solution

a $\text{Percentage mark} = \frac{43}{60} \times 100\%$

$= 71.666...\%$

$\approx 72\%$

Use $\text{Percentage} = \frac{\text{amount}}{\text{whole amount}} \times 100\%$

b $\text{Percentage} = \frac{35\text{ s}}{5\text{ min}} \times 100\%$

$= \frac{35\text{ s}}{300\text{ s}} \times 100\%$

$= 11.666...\ \%$

$\approx 11.7\%$

Change minutes to seconds so that the same units are being used.

$5\text{ min} = 5 \times 60\text{ s} = 300\text{ s}$

The **unitary method** can be used to find the whole amount (100%) when part of the amount is known. 'Unit' means 'one' and the unitary method involves finding 1% first.

The unitary method

Given a percentage of an amount, to find the whole amount:

1 find 1% of the amount by dividing by the known percentage

2 multiply by 100 to find the whole amount (100%).



Video
The unitary method

Example 6

This week, Julio received a bonus of $86, which was 9% of his weekly wage. Calculate Julio's weekly wage, correct to the nearest dollar.

Solution

9% of the wage = $86

$\therefore$ 1% of the wage = $86 ÷ 9

= $9.555...

$\therefore$ Julio's wage = $9.55... × 100

= $955.555...

$\approx$ $956

100% of the wage.

EXERCISE 3.02 ANSWERS ON P. 615

Applying percentages U F PS R C

EXAMPLE 3

1 Find:

a 15% of $85 **b** 120% of 60 kg **c** 45% of 16 m

d 2.4% of $350 **e** $5\frac{1}{4}$% of $80 **f** 12% of 95 kg

g $2\frac{1}{2}$% of $870 **h** 150% of 16 L **i** 3.75% of $2850

j $33\frac{1}{3}$% of $6540 **k** 6.25% of 25 t **l** 0.4% of $3720

2 A total of 84 600 people attended the State of Origin match between NSW and Queensland in Sydney. If 71% of the crowd supported NSW:

a what percentage supported Queensland?

b how many people supported Queensland?



Newspix/Brett Costello

3 980 students attend St Klaas' College. If 19% of them are in Year 9, calculate the number of Year 9 students.

4 Buzz Coffee advertises an extra 20% of coffee for the same price. If the normal amount is 700 g, how much extra coffee will you get for the same price?

5 Jakub's weight is now 125% of what it was a year ago. If his weight then was 85 kg, what is his weight now?

EXAMPLE 4

6 Increase:

a $455 by 10% **b** $80 by 15.5% **c** $680 by 200%

d 80 kg by 2.5% **e** 35 m by 30% **f** 120 L by 140%

7 Decrease:

a $830 by 35% **b** 120 min by 5% **c** 90 L by 12%

d 6.8 m by 25% **e** 130 kg by 18% **f** $96.50 by 10%

8 Matt buys a bike for $860 and sells it, making a profit of 5%.

a How much profit did Matt make?

b For what price did Matt sell the bike?

9 T-shirts priced at $38 are reduced by 45% at an end-of-summer sale. What is the sale price of the T-shirts?

10 Due to severe storms in summer, the price of fresh fruit and vegetables increased by 34%. How much will broccoli cost if its original price was $4.99/kg?



Foundation Standard Complex

9780170479585

11 Lucy's weekly wage of $895 increased by 5.5%. Find her new weekly wage.

12 A car dealer is offering new year sales. What will you pay for a used car marked at $16 500 if a **discount** of 12% is given?

13 Express each proportion as a percentage, correct to one decimal place if necessary. C

EXAMPLE 5

3.02

a	55 out of 80	**b**	80 out of 120	**c**	45 out of 70
d	15 out of 18	**e**	$8 out of $12	**f**	$9 out of $60
g	35 s out of 4 min	**h**	15 min out of 2 h	**i**	340 mm out of 2 m
j	3 hours out of 1 day	**k**	85 cm out of 5 m	**l**	140c out of $10

14 Selina achieved the following marks in two tests: PS R C
English: 32 out of 40
Mathematics: 47 out of 60
Convert each mark to a percentage to decide which is the better mark.

15 Travis earns $840 per week, of which $280 goes to rent. Find what percentage of Travis' wage goes to rent, correct to one decimal place.

16 Bozena earns $480 per week while she is completing Year 12 at school. If she spends $95 of it on petrol, what percentage of her wage is this, correct to one decimal place?

17 The nutritional information on a food label is shown. C

a What percentage of a 100 g serve is:

i protein?

ii carbohydrate?

b For a 100 g serve, what percentage of the total fat is saturated fat?

	Per 100 g
Energy	2105 kJ
Total Fat (of which is Saturated fat)	25 g 16 g
Carbohydrate (of which is sugars)	62 g 38 g
Protein	5 g
Salt	1 g

18 In the Netball World Cup final, Australia scored 15 goals in the second quarter. If this represented 29.4% of the goals Australia scored in the match, how many goals did Australia score in total? PS R

EXAMPLE 6

19 A jeweller charges $105 for valuing a diamond ring. If this charge represents 1.25% of the value of the ring, find this value correct to the nearest dollar.

20 A can of energy food drink contains 20% more for the same price as the standard can. If the extra amounts to 150 g, how much does the standard can contain?

21 Last year, Ben paid 28% of his salary in tax. If he paid $23 423 in tax, what was Ben's salary? Answer to the nearest dollar.

22 Indigo spent 65% of her money on rent and 50% of the remainder on food. What percentage of her money did she have left?

23 Increase $6580 by 6% and then decrease the result by 10%. Is this the same as decreasing $6580 by 4%? Show working to justify your answer. PS R

Foundation Standard Complex

24 An amount is increased by 10% and then decreased by 10%. What is the overall percentage change? PS R

25 An amount is decreased by 10% and then increased by 10%. What is the overall percentage change? PS R

Did you know?

Blood: it takes all types

The blood groups in Australia are as follows.

O		A		B		AB	
+	−	+	−	+	−	+	−
40%	9%	31%	7%	8%	2%	2%	1%

1 If Australia's population is 26 million, how many people would be in each blood group?

2 Do you know what your blood group is? How would you find out?



Shutterstock.com/ESB Professional

3.03 Profit, discounts and GST

Profit and loss

Worksheet
Profit and loss

Puzzle
Percentage profit or loss

Video
Profit and loss

Profit and loss

- When an item is sold for **more** than it costs, a **profit** is made.
- When an item is sold for **less** than it costs, a **loss** is made.
- The percentage profit and loss are usually calculated as a percentage of the **cost price**.

Example 7

The online store *Jackson's* buys notebook computers for $550 each and sells them for $680. Find:

a the profit on each computer

b the percentage profit on the cost price, correct to one decimal place.

Solution

a Profit = $680 − $550
= $130

b Percentage profit $= \frac{\text{profit}}{\text{cost price}} \times 100\%$
$= \frac{130}{550} \times 100\%$
$= 23.6363...$
$\approx 23.6\%$

Foundation Standard Complex

Discounts

Example 8

At *Jeans Minus*' end-of-season sale, jeans are discounted by 25%. What is the discount price of a pair of jeans marked at $148?

Solution

Method 1

Discount = 25% of $148

= $37

Discount price = $148 – $37

= $111

Method 2

Discount price = 75% of $148

= $111

Video
Discounts and GST

3.03

Worksheet
Discounts and special offers

Example 9

A home cinema costs $865 after a 24% discount. What was its original price, rounded to the nearest dollar?

Solution

Sale price = $865, discount = 24%

$\therefore$ 76% of the original price = $865 — 100% – 24% = 76%

$1\% = \frac{\$865}{76}$ — Using the unitary method

$= \$11.381\,578...$

$\therefore$ Original price $= \$11.381\,578... \times 100$ — To find 100%

$= \$1138.157...$

$\approx \$1138$

Video
Discounts and GST

GST

Goods and Services Tax (GST) is a tax paid to the government on most goods (items) and services that we purchase. In Australia, GST is charged at 10% of the original price and is generally included in the marked price of the good or service.

Example 10

The **selling price** of a lounge suite is $2695 with GST included. How much of this price is the GST?

Solution

Selling price = 100% + 10% — Adding 10% GST

= 110%

110% of the original price = $2695

10% GST = $2695 ÷ 11

= $245

Videos
Discounts and GST
GST

EXERCISE 3.03 ANSWERS ON P. 616

Profit, discounts and GST U F PS R

In this exercise, express all percentages correct to one decimal place and all prices to the nearest cent.

EXAMPLE 7

1 A home theatre system with a cost price of $650 is sold for $1250. Find:

a the profit **b** the percentage profit on the cost price.

2 A motorbike originally priced at $16 000 is sold for $14 500. Find:

a the loss **b** the percentage loss

3 Alla bought an old car for $3000, spent another $1500 on repairs and then sold it for $6200. Find:

a the total amount that Alla spent on the car

b the profit made when she sold the car

c her percentage profit.

4 A car costing $12 000 is sold for $8000. Calculate the percentage loss.

5 *Home Necessities* store buys cutlery sets for $120 that are then sold for $230. Calculate the percentage profit.

EXAMPLE 8

6 GIA supermarket reduces the price of some items by 35%. Find the discount price of the following items given their original prices.

a 250 g butter, $2.40 **b** 2 L milk, $3

c 1 kg cheese, $11 **d** 24 cans of soft drink, $15.99

7 Calculate the selling price of a computer marked at $760, now discounted by 15%. Select the correct answer **A**, **B**, **C** or **D**.

A $114 **B** $660 **C** $646 **D** $636

8 At a book sale, all books are discounted by 35%. What is the discount price of a book marked at $25.50?

9 A factory-seconds washing machine has its price reduced from $980 to $750. What is the percentage discount?

10 A TV priced at $1599 is sold for $1250. Find the discount as a percentage of the marked price.

11 Calculate the percentage discount for each item.

a



b



12 A jacket discounted by 40% after Christmas sells for $84. What was the original price of the jacket?

EXAMPLE 9

13 A bedroom suite reduced by 15% is sold for $2801. What was its original price? Answer to the nearest dollar.

14 A plane fare from Brisbane to Sydney has been discounted by 45%. If Amy paid $159 for a discounted ticket, what was the full price, to the nearest dollar?

15 For each item, calculate the GST payable and the final price.

a a car priced at $20 900

b a home cinema priced at $1810

c a theatre ticket priced at $145

d plumber's fees of $180

16 Given the selling price of each item with GST included, calculate the GST and the original price, correct to the nearest cent. PS R

EXAMPLE 10

a haircut: $46.20

b car repairs: $739.20

c photo frame: $31.35

d pair of jeans: $65.89

e refrigerator: $924

f piano lessons: $198 per term

17 A discount warehouse has a 30% discount on all computers with a further 5% off the discount price if you pay in cash. Carlos wants to buy a computer priced at $1899. PS R

a What is the discount price of the computer?

b Carlos pays by cash. How much will he need to pay now, to the nearest 5 cents?

c What is the total discount Carlos received?

d Calculate his total percentage discount (on the original cost).

18 Justine buys 20 boxes of paper at $29.95 a box. PS R

a What is the cost of the 20 boxes of paper?

b Justine receives a business discount of 15%. How much will she need to pay?

c Justine pays by cash and receives a further discount of 5%. How much will she now pay, to the nearest 5 cents?

d Find the amount Justine would pay if she received a single discount of 20% on the cost of the paper.

e Is the single discount of 20% the same as the two successive discounts of 15% and 5%? Give reasons for your answer.

Quiz
Mental skills 3A

Mental skills 3A: Maths without calculators ANSWERS ON P. 616

Squaring a number ending in 5, 1 or 9

The square of a number ending in 5 always ends in 25.
For example, $35^2 = 1225$ and $105^2 = 11\,025$.
A mental calculation trick requires three easy steps:

- delete the 5 from the number
- multiply the remaining number by the next consecutive number
- write '25' at the end of the product.

1 Study each example.

a To calculate 35^2:

- deleting the 5 from 35 leaves 3
- multiply 3 by the next consecutive number: $3 \times 4 = 12$
- write '25'at the end: 1225

$35^2 = 1225$

b To calculate 105^2:

- deleting the 5 from 105 leaves 10
- multiply 10 by the next consecutive number: $10 \times 11 = 110$
- write 25 at the end: 11 025

$105^2 = 11\,025$

2 Now calculate each square number.

a	25^2	**b**	55^2	**c**	45^2	**d**	65^2	**e**	75^2	**f**	2.5^2
g	4.5^2	**h**	205^2	**i**	6.5^2	**j**	7.5^2	**k**	275^2	**l**	345^2

The square of a number ending in 1 always ends in 1.

For example, $41^2 = 1681$ and $71^2 = 5041$.

A mental calculation trick requires three steps.

- Round the number to the nearest 10 (by subtracting 1) to make a new number
- Square the new number
- Add to your answer the new number and its next consecutive number

3 Study each example.

a To calculate 41^2:

- Round 41 down to 40.
- Square 40: $40^2 = 1600$
- Add 40 and 41: $1600 + 40 + 41 = 1681$

$41^2 = 1681$

b To calculate 71^2:

- $70^2 = 4900$ Rounding down and squaring
- $4900 + 70 + 71 = 5041$

$71^2 = 5041$

4 Now calculate each square number.

a	21^2	**b**	101^2	**c**	31^2	**d**	91^2	**e**	5.1^2
f	81^2	**g**	61^2	**h**	201^2	**i**	2.1^2	**j**	9.1^2

The square of a number ending in 9 also ends in 1.

For example, $29^2 = 841$ and $99^2 = 9801$.

A mental calculation trick requires three steps:

- round the number to the nearest 10 (by adding 1) to make a new number
- square the new number
- subtract from your answer the new number and its *previous* consecutive number

5 Study each example.

a To calculate 29^2:

- round up to 30
- square 30: $30^2 = 900$
- subtract 30 and 29:

$900 - 30 - 29 = 841$

$29^2 = 841$

b To calculate 99^2:

- round up to 100
- square 100: $100^2 = 10\,000$
- $10\,000 - 100 - 99 = 9801$

$99^2 = 9801$

6 Now calculate each square number.

a	59^2	**b**	69^2	**c**	89^2	**d**	19^2	**e**	109^2
f	5.9^2	**g**	6.9^2	**h**	119^2	**i**	39^2	**j**	8.9^2

Simple interest 3.04

Banks, credit unions and other financial institutions reward investors by paying them interest on their savings or investments. Conversely, they charge borrowers by making them pay interest on their loans.



Worksheet
Simple interest

- The original amount of money invested or borrowed is called the **principal**.
- **Interest** is calculated as a percentage of the principal.
- This percentage is called the **interest rate**, usually written as a rate **per annum** ('per year'), abbreviated 'p.a.'
- **Simple interest** (or **flat rate interest**) is interest calculated simply on the original principal.

Example 11

Find the simple interest earned on an investment of $18 000 at 3.5% p.a. for 5 years.

Solution

Principal = $18 000, interest rate = 3.5% p.a., term = 5 years.

Term means the amount of time the investment or loan has occurred

Interest = $3.5\% \times \$18\,000 \times 5$ Over 5 years.

$= \$3150$

The simple interest formula

$I = Prn$, where:

I is the simple interest

P is the principal

r is the interest rate per year, expressed as a decimal

n is the number of years

Applying this to the above example,

$P = \$18\,000$, $r = 3.5\% = 0.035$, $n = 5$ years

$I = Prn$

$= \$18\,000 \times 0.035 \times 5$

$= \$3150$

Video
Simple interest

Example 12

Find the simple interest on:

a $8620 at 2.4% p.a. for 7 months

b $5600 at 6.25% p.a. for 220 days

Solution

a $P = \$8620$, $r = 2.4\% = 0.024$,

$n = 7$ months $= \frac{7}{12}$ years n must be in years

$I = Prn$

$= \$8620 \times 0.024 \times \frac{7}{12}$

$= \$120.68$

b $P = \$5600$, $r = 6.25\% = 0.0625$,

$n = 220$ days $= \frac{220}{365}$ years

$I = Prn$

$= \$5600 \times 0.0625 \times \frac{220}{365}$

$= \$210.9589...$

$\approx \$210.96$ rounded to the nearest cent

9780170479585

Example 13

After 2 years, an investment of \$1560 has earned \$87.36 in simple interest.

What is the annual interest rate?

Solution

$I = \$87.36$, $P = \$1560$, $n = 2$ years

$I = Prn$

$\$87.36 = \$1560 \times r \times 2$

$\$87.36 = \$3120r$

$r = \frac{\$87.36}{\$3120}$

$= 0.028$

$= 2.8\%$

$\therefore$ Annual interest rate $= 2.8\%$

Video
Simple interest

3.04

Example 14

For how long will \$12 000 need to be invested to earn \$1350 if the flat rate of interest is 4.5% p.a.?

Solution

$I = \$1350$, $P = \$12\,000$, $n = 4.5\% = 0.045$

$I = Prn$

$1350 = 12\,000 \times 0.045 \times n$

$1350 = 540n$

$n = \frac{1350}{540}$

$= 2.5$ years

EXERCISE 3.04 ANSWERS ON P. 616

Simple interest U F PS R

In this exercise, give all money answers correct to the nearest cent.

1 Calculate the simple interest earned on each investment. EXAMPLE 11

a \$18 000 at 7% p.a. for 5 years

b \$5280 at 2.85% p.a. for 2 years

c \$3000 at 5.5% p.a. for 4 years

d \$9450 at $2\frac{1}{2}\%$ p.a. for 3 years

e \$6800 at 6.2% p.a. for 6 years

f \$12 500 at 7.3% p.a. for 2 years

2 Steffi invests \$17 000 at 6.45% p.a. simple interest for 4 years. To what final value will her investment grow?

3 Calculate the simple interest for each situation. EXAMPLE 12

a \$2100 invested at 6% p.a. for 9 months

b \$3450 invested for 5 months at 3.25% p.a.

c \$10 000 invested at 6.2% p.a. for 150 days

d \$4000 borrowed for 14 months at 6.55% p.a.

e \$280 000 borrowed at 8.3% p.a. for 30 weeks

f \$18 400 borrowed for 160 days at 14.7% p.a.

EXAMPLE 13

4 Find the annual interest rate if:

a an investment of $3000 earns $756 simple interest after 4 years

b an investment of $17 630 earns $2715.02 simple interest after 2 years

c a flat rate loan of $9000 is charged $2115 after 5 years.

EXAMPLE 14

5 For how long will $12 350 need to be invested to earn $1778.40 if the flat rate of interest is 2.4% p.a.? PS R

6 How many months will it take for an investment of $22 000 to earn $12 650 in interest if the interest rate is 6.9% p.a.?

7 How many days will it take for $15 600 to earn $436.97 in interest if the flat rate of interest is 7.1%?

8 William took 2 years to pay off a flat rate loan of $7000. His total loan repayments amounted to $8400. Calculate:

a the interest charged **b** the interest rate

9 Simone and Jeff borrowed $4000 to build a garden in their backyard. They repaid the loan at $128 per month for 4 years. PS

a Calculate the total amount they paid for the garden.

b How much interest did they pay on the loan?

c Calculate the flat rate of interest charged.



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3.05 Ratios and rates

Skillsheet
Ratios

Interactives
Ratio and proportion
Proportion playground
Unit rates

A **ratio** compares quantities of the same type measured in the same units.

It is written in the form '$a : b$' (pronounced 'a to b'), where a and b are numbers.

Simplifying ratios

To **simplify a ratio**, keep dividing each term by the same number, preferably a large number such as their highest common factor (HCF), until each term is as small as possible.

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Example 15

Simplify each ratio.

a 36 : 48 : 18

b 425 mL to 5 L

Solution

a $36 : 48 : 18 = \frac{36}{6} = \frac{48}{6} = \frac{18}{6}$
$= 6 : 8 : 3$

b $425 \text{ mL} : 5 \text{ L} = 425 \text{ mL} : 5000 \text{ mL}$
$= 425 : 5000$
$= \frac{425}{25} : \frac{5000}{25}$
$= 17 : 200$

Ratio problems can be solved using **equivalent ratios** or the **unitary method**.
The unitary method requires finding one part first.



Video
Ratio problems

Example 16

During summer, the ratio of jeans to shorts sold at a beachside store was 3 : 10.
If 240 pairs of jeans were sold, how many pairs of shorts were sold?

Solution

Method 1: Equivalent ratios

Jeans : shorts = 3 : 10 = 240 : _____

Since $3 \times 80 = 240$,

Number of shorts $= 10 \times 80 = 800$

Method 2: Unitary method

Jeans : shorts = 3 : 10

Since 240 jeans were sold, 3 parts = 240

$\therefore$ 1 part $= 240 \div 3 = 80$

$\therefore$ Number of shorts $= 80 \times 10 = 800$

Dividing a quantity in a given ratio

- Find the total number of parts
- If using the **unitary method**, find the size of one part, then multiply to calculate each share
- If using the **fraction method**, calculate fractions of the quantity to find each share
- Check that the shares add up to the original quantity

Video
Dividing a quantity in a given ratio

Example 17

Divide a cash prize of $420 between Elyse and Henry in the ratio 3 : 4.

Solution

Method 1: Unitary method

Total number of parts = 3 + 4 = 7

∴ 7 parts = $420

1 part = $420 ÷ 7 = $60

∴ Elyse's share = 3 × $60 = $180

Henry' share = 4 × $60 = $240

Checking: $180 + $240 = $420

Method 2: Fraction method

Total number of parts = 3 + 4 = 7

∴ Elyse's share = $\frac{3}{7} \times \$420 = \$180$

Henry' share = $\frac{4}{7} \times \$420 = \$240$

Checking: $180 + $240 = $420

Simplifying rates

A **rate** compares 2 quantities of different types measured in different units.

It measures how one quantity changes with another quantity. It is written in the form 'a/b' (pronounced 'a per b'), where a and b are units of measurement.

Example 18

Write each statement as a rate.

a Travelling 600 km in 12 hours.

b A phone call costs $6.45 for 15 minutes.

Solution

a 600 km in 12 h = 600 ÷ 12
= 50 km/h

b $6.45 for 15 min = $6.45 ÷ 15
= $0.43/min

Video
Rate problems

Rate problems

Problems involving rates can be solved usually by multiplying or dividing.
The following strategy may help.

- Write the units of the rate x/y as a fraction: $\frac{x}{y}$
- To find the quantity in the numerator, x, **multiply** by the rate
- To find the quantity in the denominator, y, **divide** by the rate

Worksheet
Rates exercise

Example 19

A car travels at 92 km/h.

a How far will it travel in $4\frac{1}{2}$ hours?

b How long will it take to travel 800 km? Answer correct to one decimal place.

Solution

a Write the units of the rate as a fraction: km/h $= \frac{\text{km}}{\text{h}}$

To find the distance (km), multiply by the rate.

Distance $= 4\frac{1}{2} \times 92$

$= 414$ km

b To find the time (h), divide by the rate.

Time $= 800 \div 92$

$= 8.6956...$

≈ 8.7 h

EXERCISE 3.05 ANSWERS ON P. 616

Ratios and rates U F PS R

1 Simplify each ratio.

a	48 : 28	**b**	15 : 36	**c**	125 : 60	**d**	84 : 56
e	124 : 62	**f**	$\frac{3}{5} : \frac{1}{4}$	**g**	0.68 : 0.24	**h**	1.25 : 4.5
i	$5 : \frac{4}{5}$	**j**	$\frac{5}{8} : \frac{2}{3}$	**k**	4.8 : 0.64	**l**	15 : 18 : 6
m	27 : 36 : 9	**n**	120 : 20 : 80	**o**	$\frac{5}{6} : 3$	**p**	$2\frac{1}{2} : 5$

EXAMPLE 15

2 Simplify each ratio.

a	75c : \$4	**b**	5 kg : 800 g	**c**	2 hours : 45 min
d	600 mm : 2 m	**e**	8 hours : 1 day	**f**	8 mm : 4 cm
g	300 mL : 2 L	**h**	\$3.50 : 25c	**i**	6 min : 45 s

3 The ratio of Ken's age to his daughter Jess' age is 10 : 3. If Ken is 30, How old is Jess?

EXAMPLE 16

4 Bella and Suri invest in a business in the ratio 3 : 5. If the larger investment is \$520 000, find the amount of the smaller investment.

5 A survey of car buyers found that they purchased cars with colours in the ratio of white : grey : blue : other colour = 8 : 5 : 11 : 10.

A car dealer ordered 20 cars in colours other than white, grey and blue. How many white cars should be ordered?

6 A concrete mixture of gravel to sand to cement of 4 : 3 : 1 is needed for strong foundations. How much of each is needed to make 40 cubic metres of concrete? PS R

EXAMPLE 17

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7 At a busy intersection involving a red-light camera, the ratio of cars running a red light to cars stopping was 1 : 79. If 90 000 cars passed through the intersection last week, how many ran the red light?



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8 Shahid and Bridie won $60 000 in a lottery. If they share the prize in the ratio 23 : 27, how much does each person receive?

9 An alloy consists of nickel and copper in the ratio 4 : 7. If the alloy weighs 3.41 kg, how much copper was used? Select the correct answer **A**, **B**, **C** or **D**.

A 0.21 kg **B** 0.85 kg **C** 1.24 kg **D** 2.17 kg

EXAMPLE 18

10 Write each statement as a rate using the units in brackets. Round answers to one decimal place if needed.

a 540 km in 8 h (km/h)
b 5 kg for $42 ($/kg)
c 2800 words in 50 min (words/min)
d 14 L for 170 km (km/L)
e 200 m in 26 s (m/s)
f 5 kg of seed for 120 m^2 (g/m^2)
g 40 days to walk 900 km (km/day)
h $215 for 3 hours ($/h)

EXAMPLE 19

11 A truck maintains an average speed of 65 km/h. Calculate how far it will travel:

a in $2\frac{1}{2}$ h
b from 6:30 a.m. to 3:00 p.m.

12 Walid earns $220 for an 8-hour day. PS R

a Express this pay as an hourly rate ($/h).
b How much would he earn in a 40-hour week?
c How many days will it take him to earn $1760?

13 James' heart beats at 78 beats/min.

a How many times will it beat in one hour?
b How long will it take to beat 1000 times? Answer to the nearest second.

14 Gold jewellery is classified according to its gold content. The ratio of gold to other metals is given in carats. Pure gold is 24 carats, so 10-carat gold has gold mixed with other metals in the ratio 10 : 14. PS R

a Write the ratio of gold to other metals in:
i a 9-carat bracelet
ii an 18-carat ring

b Nami purchased a 14-carat gold necklace with a mass of 50 g. How much gold is in the necklace, correct to one decimal place?

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15 A utility van has a fuel consumption of 11.6 L/100 km. Calculate how much fuel it will use for a trip of:

a 640 km **b** 58 km **c** 140 km

16 Calculate the fuel consumption, in L/100 km, of a Toyota Prius hybrid that uses 30.6 L to travel 900 km.

17 Birth rates are given as births/1000 people. A city's birth rate is 16 births/1000 people. If there were 520 births in the city last year, what was its population?

18 David claims that 5 kg of Greenie grass seed will plant an area of 1200 m^2. How much grass seed is needed for an area of 60 m^2? Select **A**, **B**, **C** or **D**.

A 0.25 kg **B** 0.4 kg **C** 1.44 kg **D** 4 kg

19 **a** Indonesia has a population of 278 million and an area of 1 919 440 km^2. Calculate its population density in people/km^2, correct to one decimal place.

b Australia has a much smaller population of 26.6 million but a much larger area of 7 692 000 km^2. Calculate its population density, correct to one decimal place.

c If Australia was as densely populated as Indonesia, what would its population be? Answer to the nearest hundred.

20 A large pump takes 4 minutes to fill a tank. A smaller pump takes 10 minutes to fill the same tank. How long, in minutes and seconds, will it take to fill the tank, if both pumps are turned on at the same time?

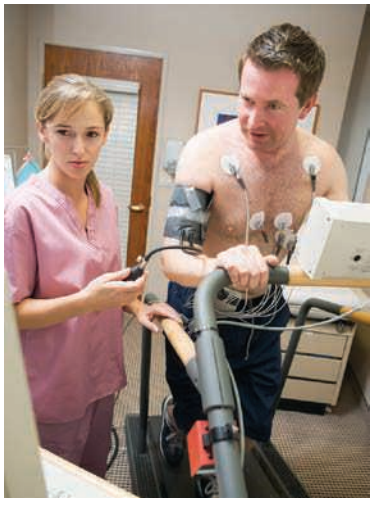
Did you know?

The beat of life

The heart of an unfit person works harder whether at rest, during activity or even while recovering from activity. A fit person's heart copes with activity better and its beat returns to normal sooner.

	Heartbeats per minute		
	Fit	Unfit	Difference
Before activity	65	80	15
During activity	95	135	40
3 minutes after	80	115	35
6 minutes after	70	100	30
9 minutes after	65	95	30

What patterns do you notice in the differences in heart rates between fit and unfit persons?



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3.06 Converting rates

It is often necessary to convert rates from one set of units to another, for example, from kilometres/hour to metres/second, or from millilitres/minute to litres/day.

Video
Converting rates 2

Example 20

Convert:

a 80 L/kg to L/g **b** 5 m/s to km/h **c** 2 mL/min to L/day

Solution

a $80 \text{ L/kg} = \frac{80}{1000} \text{ L/g}$ 1 kg = 1000 g
$= 0.08 \text{ L/g}$

b $5 \text{ m/s} = 5 \times 3600 \text{ m/h}$ 1 h = 3600 s
$= 18\,000 \text{ m/h}$
$= \frac{18\,000}{1000} \text{ km/h}$ 1 km = 1000 m
$= 18 \text{ km/h}$

c $2 \text{ mL/min} = 2 \times 60 \text{ mL/h}$ 1 h = 60 min
$= 120 \text{ mL/h}$
$= 120 \times 24 \text{ mL/day}$ 1 day = 24 h
$= 2880 \text{ mL/day}$
$= \frac{2880}{1000} \text{ L/day}$ 1 L = 1000 mL
$= 2.88 \text{ L/day}$

EXERCISE 3.06 ANSWERS ON P. 616

Converting rates U F PS R

EXAMPLE 20

1 Convert:

a 5 m/s to m/h **b** 30 m/s to m/h **c** \$10/kg to \$/g
d 20 mL/min to mL/h **e** 40 mL/min to mL/h **f** 5 mL/s to mL/h
g 5 kg/m^2 to kg/ha **h** 2.5 tonnes/day to kg/day **i** \$750/week to \$/year
j 0.5 km/min to km/h **k** 15 sheep/h to sheep/day **l** \$60/day to \$/week

2 Convert (round to one decimal place if needed):

a 5 m/s to km/h **b** 35 cm/s to m/h **c** 10 cents/g to \$/kg
d 40 mL/min to L/h **e** 40 L/min to mL/s **f** 5 mL/s to L/day
g 25 g/m^2 to kg/ha **h** 2.5 tonnes/h to kg/day **i** 110 km/h to m/s
j 5 kg/m to g/cm **k** 8 km/L to m/mL **l** 70c/min to \$/h
m 45 mm/s to km/h **n** 0.03 g/mL to kg/kL **o** 0.45 kL/h to mL/s

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3 Convert each speed into kilometres per hour. Round your answers to two decimal places.

a An African cheetah was measured running at 27 m/s.

b A German peregrine falcon dived at 97 m/s.

c A Tanzanian snake can travel at a speed of 3.3 m/s.

d A sailfish off the coast of Florida was estimated to swim 30 m/s.

e A racing cyclist rode at 23 m/s.

4 If Nate's reaction time is 0.9 seconds, how far will his car travel in this time if its speed is: PS R

a 60 km/h? **b** 80 km/h? **c** 110 km/h?

Mental skills 3B: Maths without calculators ANSWERS ON P. 616

Quiz
Mental skills 3B

Estimating square roots

We can use the square numbers to estimate square roots.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1	4	9	16	25	36	49	64	81	100	121	144

1 Study each example.

a Estimate $\sqrt{7}$.

7 is between the square numbers 4 and 9 (2^2 and 3^2 respectively).

Because 7 is closer to 9, $\sqrt{7}$ is closer to 3.

An estimate is $\sqrt{7} \approx 2.6$. (Actual answer = 2.645...)

b Estimate $\sqrt{55}$.

55 is between the square numbers 49 and 64 (7^2 and 8^2 respectively).

Since 55 is closer to 49, $\sqrt{55}$ is closer to 7.

An estimate is $\sqrt{55} \approx 7.4$. (Actual answer = 7.416...)

2 Now estimate each square root correct to one decimal place.

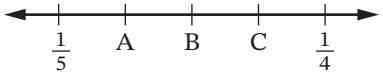
a $\sqrt{8}$ **b** $\sqrt{18}$ **c** $\sqrt{28}$ **d** $\sqrt{111}$

e $\sqrt{80}$ **f** $\sqrt{31}$ **g** $\sqrt{12}$ **h** $\sqrt{65}$

i $\sqrt{75}$ **j** $\sqrt{29}$ **k** $\sqrt{40}$ **l** $\sqrt{126}$

Power plus ANSWERS ON P. 616

1 What fractions do A, B and C represent on this number line if the intervals between the fractions are equal?



2 Deepti sold a car for $14 700, making a loss of 30% on its original price. Find the original price.

3 A store marks up the price of a lounge suite by 80%. What percentage discount would bring the price of the lounge suite back to its original price?

4 Ineke went to a supermarket and bought 5 kg of potatoes. If 3 kg of potatoes cost $4.99, how much change would Ineke receive from $20?

5 2 L of water is to be shared equally between 3 people.

a Calculate how much water each person would receive (correct to 3 decimal places).

b Explain why 3 decimal places would be too accurate for practical purposes.

6 Melinda bought a pair of shoes that were reduced by $\frac{1}{3}$. If the shoes sold for $72, what was their original price?

7 After training for 6 months, Owen improved his time for running 100 m by 8%. If his new time is 10.2 seconds, what was his old time? Answer correct to one decimal place.

8 At a local football game, the ratio of men to children was 5 : 8 and the ratio of women to children was 7 : 10. What is the ratio of men to women?

9 Hannah, Nicola and Olivia share an amount of money in the ratio 2 : 3 : 5. If Nicola gives $35 to Hannah, both of them will have an equal amount of money. How much money will Olivia receive?

10 James and Jordan each drove their cars to the beach. James drove his car at a speed of 40 km/h for half the distance and at 60 km/h for the other half. Jordan drove at 50 km/h for the entire distance. Who arrived first, James or Jordan? Give reasons for your answer.

11 **a** How long will it take for an investment of $5000 to double if it is invested at 5% p.a. simple interest?

b How long will it take for an investment of $10 000 to double at the same interest rate?

CHAPTER 3 REVIEW

Quiz
Language of maths 3

Puzzle
Numbers crossword

Language of maths

convert	cost price	decimal	decrease
discount	flat rate interest	fraction	GST
increase	interest rate	irrational	loss
per	per annum (p.a.)	percentage	principal
profit	rate	ratio	recurring
selling price	simple interest	terminating	unitary method

1 What is another name for **flat rate interest**?

2 What is the product of a fraction and its reciprocal?

3 Find a non-mathematical meaning for:

a reciprocal **b** recurring **c** terminating.

4 What is the difference between the selling price and the cost price called when the selling price is higher?

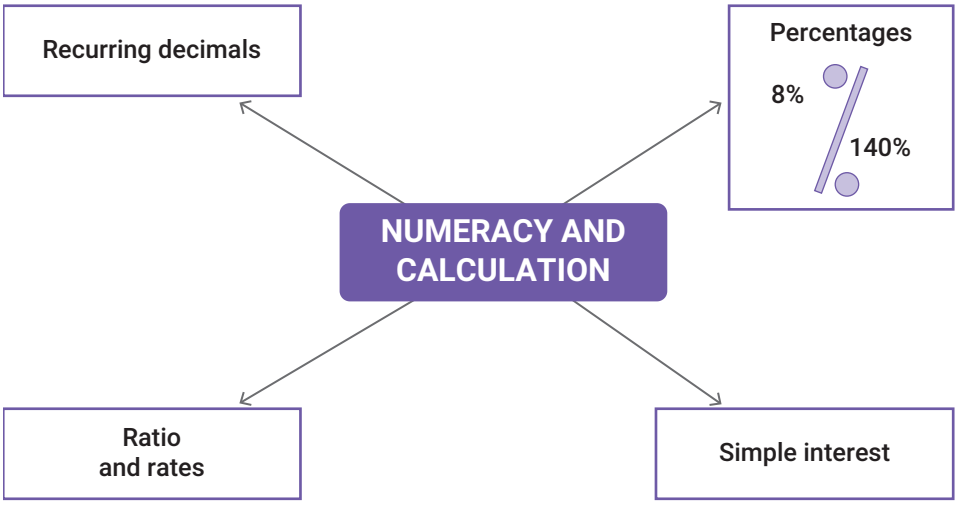
5 What does 'annum' mean, as in 'per annum'?

6 Which method involves finding the value of one part when given the value of the whole amount or several parts?

Topic summary

Write 10 questions (with solutions) that could be used in a test for this chapter. Include some questions that you have found difficult to answer.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Numeracy and calculation (Advanced)

Quiz
Test yourself 3

TEST YOURSELF 3

ANSWERS ON P. 616

3.01 **1** Convert $\frac{4}{11}$ to a recurring decimal.

PATH

3.01 **2** Convert each recurring decimal to a simple fraction.

a $0.\dot{8}$ **b** $0.\dot{2}\dot{7}$ **c** $0.1\dot{6}$ **d** $2.\dot{6}4\dot{2}$

3.02 **3** **a** Increase 120 L by 35%. **b** Decrease $25 by 12.5%

3.02 **4** What percentage is (correct to one decimal place):

a 5 minutes of 1 hour? **b** 75c out of $4.50?

3.03 **5** A block of land was purchased for $250 000. Four years later, it was sold for $630 000. Calculate:

a the profit **b** the profit as a percentage of cost price.

3.03 **6** At a sale, Garrett and Kay bought a cutlery set, saving $150 on the original price. If this discount represented 55% of the price, what was the price of the cutlery?

3.04 **7** Calculate, to the nearest cent, the simple interest on $3890 borrowed for 7 months at 2.4% p.a.

3.04 **8** Calculate the annual simple interest rate if an investment of $7000 earns $1820 in interest over 2 years.

3.05 **9** Bill and Ben win $800 000 in Lotto. If they agree to share the prize in the ratio of 15 : 17, how much will each person receive?

3.05 **10** A truck travels 581 km in 7 hours.

a Calculate its average speed in km/h.

b How far does the truck travel in 4 hours?

c How long will it take to travel 1000 km? Answer correct to one decimal place.

3.05 **11** A farmer spreads fertiliser at the rate of 15 kg/ha. Calculate:

a the amount of fertiliser needed to cover an area of 26 000 m^2.

b the area, in hectares, which will be covered by 450 kg of fertiliser.

3.06 **12** Convert:

a 0.8 mL/s to mL/h **b** 140 g/m^2 to kg/m^2

c 30 km/h to m/s **d** 345.6 m/day to cm/min

3.05 **13** The ratio of Zoe's savings to Lucy's savings was 9 : 4. After Zoe spent $81, the ratio became 3 : 2. What was the value of Lucy's savings?

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PRACTICE SET 1

ANSWERS ON P. 617

1 Convert each fraction to a decimal and state whether it is recurring or terminating. 3.01

a $\frac{5}{8}$ **b** $\frac{7}{13}$ **c** $\frac{8}{9}$ **d** $4\frac{3}{4}$

2 Simplify each expression. 1.01

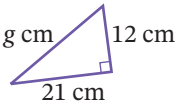
a $7 + 5m - 9$ **b** $8p - 3m + 12p + 4m$ **c** $5n^2 - 3n - 2n^2 - 7n$

3 Select the surds from this list of numbers. 2.02

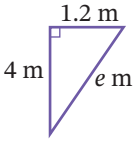
$\sqrt{98}$ $\sqrt{9}$ $\sqrt{225}$ $\sqrt{160}$ $\sqrt{36}$ $\sqrt{0.25}$ $\sqrt{52}$

4 Find, correct to one decimal place, the value of each variable. 2.01

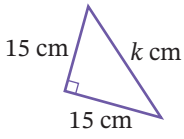
a



b



c



PATH

5 Convert each recurring decimal to a simple fraction. 3.01

a $0.\dot{4}\dot{2}$ **b** $0.2\dot{6}$ **c** $0.\dot{1}8\dot{0}$ **d** $3.\dot{2}1\dot{6}$

6 Simplify each surd. 2.03

a $\sqrt{75}$ **b** $\left(3\sqrt{7}\right)^2$ **c** $2\sqrt{28}$ **d** $\sqrt{162}$

7 Expand each expression. 1.09

a $(5y - 7)(5y + 7)$ **b** $(2g + 3m)(2g - 3m)$ **c** $\left(4u - \frac{1}{u}\right)\left(4u + \frac{1}{u}\right)$

8 Expand each expression. 1.06

a $(p + 4)(p - 8)$ **b** $(2n + 1)(3n + 5)$ **c** $(7 - y)(7 + y)$

PATH

9 Expand each expression. 1.10

a $(5a + 3)^2$ **b** $(3r + 4)(3r - 4)$ **c** $(x - 3)(x + 3) - (x - 3)^2$

10 Simplify each expression. 2.04

a $10\sqrt{2} - 3\sqrt{5} + 3\sqrt{2}$ **b** $\sqrt{18} - \sqrt{8}$ **c** $\sqrt{50} + 5\sqrt{32}$

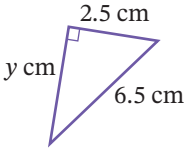
11 Simplify each expression. 2.05

a $\sqrt{3} \times \sqrt{18}$ **b** $3\sqrt{2} \times 5\sqrt{2}$ **c** $4\sqrt{5} \times \left(-\sqrt{8}\right)$

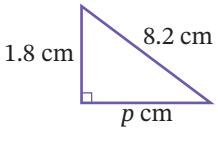
d $\sqrt{27} \div \sqrt{3}$ **e** $-8\sqrt{15} \div 2\sqrt{5}$ **f** $\frac{-7\sqrt{12}}{\sqrt{48}}$

12 Find the value of each variable. 2.06

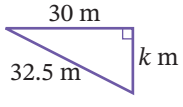
a



b



c



Foundation Standard Complex

1.07

13 Which expression is $16mn + 24m^2$ completely factorised? Select the correct answer **A**, **B**, **C** or **D**.

A $4(4n + 6m)$ **B** $8m(2n + 3m)$ **C** $4m(4n + 6m)$ **D** $m(16n + 24m)$

1.07

14 Factorise each expression.

a $3p - 15$ **b** $5(m + 4) - t(m + 4)$

c $-8x^2y - 12xy$ **d** $2ky^2 - 3ky - 2y + 3$

PATH

1.14

15 Factorise each expression.

a $36y^2 - 49x^2$ **b** $m^2 - 5mn + 3m - 15n$

c $8m^2 - 50$ **d** $2ky^2 - 3ky - 2y + 3$

1.08

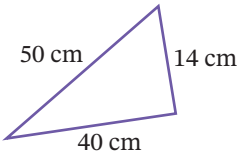
16 Expand each perfect square.

a $(3m - 5)^2$ **b** $(2y + 9)^2$ **c** $\left(h - \frac{4}{h}\right)^2$ **d** $(4p + 3w)^2$

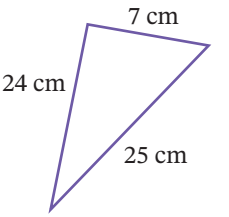
2.08

17 Test whether each triangle is right-angled.

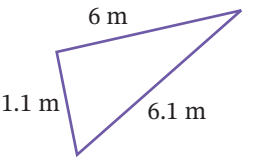
a



b



c



3.02

18 Calculate:

a 170% of \$60 **b** 14% of \$250 **c** $3\frac{1}{2}$% of \$670

3.03

19 Paba bought 55 L of petrol at 189.9 cents per litre.

a How much did Paba pay (to the nearest cent)?

b How many litres can be bought for \$60? Answer to one decimal place.

PATH

1.12

20 Factorise each expression.

a $y^2 - 2y - 63$ **b** $k^2 - 7k + 10$ **c** $w^2 + 5w - 14$ **d** $m^2 - 17m + 72$

1.13

21 Factorise each expression.

a $5k^2 + 14k - 3$ **b** $3y^2 + y - 24$ **c** $4h^2 - 7h - 15$ **d** $8w^2 - 38w + 35$

e $6y^2 + 25y + 14$ **f** $12m^2 - 31m + 20$ **g** $48a^2 + 22a - 15$ **h** $36x^2 + 73x + 35$

1.15

22 Simplify each expression.

a $\frac{7}{7k - 21}$ **b** $\frac{x + 3}{x^2 - 9}$ **c** $\frac{5y + 25}{30}$ **d** $\frac{d^2 - 49}{d^2 + 2d - 35}$

1.15

23 Simplify each expression.

a $\frac{n + 1}{n^2 - n} - \frac{2}{n^2 - 1}$ **b** $\frac{5y + 20}{9} \times \frac{3y}{y + 4}$ **c** $\frac{k^2 + 4k}{k - 3} \div \frac{4k}{k^2 - 4k + 3}$

d $\frac{w^2 + 5w - 24}{w^2 - 7w + 10} \times \frac{w^2 - 4}{w^2 + 5w + 6}$ **e** $\frac{4}{k^2 - 4k - 5} + \frac{2}{k^2 - 11k + 30}$

Foundation Standard Complex

9780170479585

PATH

24 Factorise each expression.

a $25 - 9y^2 - 30 + 18y$ **b** $28m^2 - 5m - 12$ **c** $4k^4 - 64$

25 Simplify each expression.

a $2a^2 \times 4a$ **b** $5 \times 4p \times 2pd$ **c** $9b^2d \times 3bd$ **d** $\frac{3}{4}xw^2 \times (-12xw)$

26 Simplify each expression.

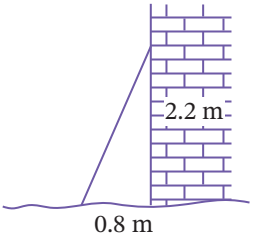
a $\frac{21mn}{7n}$ **b** $36x \div 12y$ **c** $\frac{w^2v^3}{2vw}$ **d** $35a^3d^2 \div 15a^2d^4$

27 A dishwasher costing \$525 is sold for \$699.95. Calculate the percentage profit on the cost price, correct to one decimal place.

28 Simplify:

a 160 : 96 **b** 3 h : 45 min **c** 1275 words in 15 minutes (as words/min)

29 The foot of a ladder is placed 0.8 m from the base of a wall. If the ladder reaches 2.2 m up the wall, calculate, correct to one decimal place, the length of the ladder.



30 **a** Increase 85 kg by 30%

b Decrease \$6.40 by 18%

31 Expand each expression.

a $8(5 - x)$ **b** $3h(2 - 7h)$ **c** $9c(2d - 3c)$

32 At a sale, the price of a TV was reduced by 30%. Find the original price of the TV if its sale price is \$339.50.

33 Expand and simplify each expression.

a $2(5w - 3) + 4(1 + 3w)$ **b** $3(1 - 4k) - 2(2k + 1)$ **c** $4(3t - 4) - 7(1 - 2t)$

34 Divide \$450 between Klaas, David and Sarah in the ratio 3 : 5 : 7.

35 The speed of a spacecraft leaving the Earth's gravitational pull must reach 11.2 km/s. Express this speed in km/h.

36 Simplify each expression.

a $\frac{9d}{11} - \frac{3d}{11}$ **b** $\frac{7k}{8} + \frac{11k}{8}$ **c** $\frac{3m}{5} + \frac{2m}{3}$ **d** $\frac{2y}{7} - \frac{3h}{4}$

37 Simplify each expression.

a $\frac{a}{6} \div \frac{m}{5}$ **b** $\frac{2d}{w} \times \frac{5w^2}{6m}$ **c** $\frac{3x}{5} \div \frac{9x^2}{10}$ **d** $\frac{a}{4m} \times \frac{2d}{5y} \div \frac{2y}{3m}$

38 A rectangle has side lengths of 5 cm and 7 cm. Find the length of a diagonal as a surd.

39 A discount of \$75 represents 4% of the marked price. What is the marked price?

40 Expand and simplify each expression.

a $4(2 + 3y - x^2) - 5(y - 1)$ **b** $-4(2 + 3y - x^2) + 5(y - 1)$

PATH: 1.14, 1.02, 1.02, 3.03, 3.05, 2.07, 3.02, 1.05, 3.03, 1.05, 3.05, 3.06, 1.03, 1.04, 2.07, 3.03, 1.05

Foundation Standard Complex